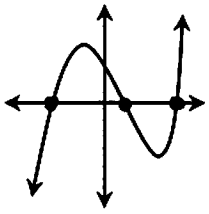
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| Main Ideas/Questions | Notes/Examples |
|---|---|
| **ZEROS**  | • The x - intercepts of a function are also called zeros.<br>• A polynomial function of degree $n$ can have at most n real zeros.<br>• An **even degree** polynomial has an even number of real zeros. (or no real zeros)<br>• An **odd degree** polynomial has an odd number of real zeros. |
| **Linear Factors & Solutions** | • The Solutions to the equation $f(x)=0$ are the zeros of the function. They are also called roots.<br>• If $x = c$ is a solution to the equation, then $(x-c)$ is a linear factor of the polynomial function.<br>• Linear factors can be found by factoring the polynomial.<br>**For example:** Given $f(x) = x^3 + 5x^2 - 24x$<br>**a)** What are the linear factors? $f(x)=x(x+8)(x-3)$<br>**b)** What are the zeros? $x=\{-8, 0, 3\}$ |
| **More Examples** | **Directions:** Find the zeros of the functions below by factoring. |

**1.** $f(x) = x^3 + 3x^2 - 10x$

$f(x) = x(x^2+3x-10)$

$f(x) = x(x+5)(x-2)$

$\boxed{x=\{-5, 0, 2\}}$

**2.** $f(x) = 2x^3 - 5x^2 - 3x$

$f(x) = x(2x^2-5x-3)$

$f(x) = x(2x+1)(x-3)$

$\boxed{x=\{-\frac{1}{2}, 0, 3\}}$

**3.** $f(x) = x^4 - 5x^2 + 4$

$f(x) = (x^2-4)(x^2-1)$

$f(x) = (x+2)(x-2)(x+1)(x-1)$

$\boxed{x=\{\pm 1, \pm 2\}}$

**4.** $f(x) = 9x^4 - 37x^2 + 4$

$f(x) = (9x^2-1)(x^2-4)$

$f(x) = (3x+1)(3x-1)(x+2)(x-2)$

$\boxed{x=\{\pm \frac{1}{3}, \pm 2\}}$

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**5.** $f(x) = x^3 + 5x^2 - 9x - 45$

$f(x) = x^2(x+5) - 9(x+5)$

$f(x) = (x^2 - 9)(x+5)$

$f(x) = (x+3)(x-3)(x+5)$

$x = \{-5, \pm 3\}$

of

**6.** $f(x) = 20x^3 - 12x^2 - 5x + 3$

$f(x) = 4x^2(5x-3) - 1(5x-3)$

$f(x) = (4x^2 - 1)(5x-3)$

$f(x) = (2x+1)(2x-1)(5x-3)$

$x = \{\pm\frac{1}{2}, \frac{3}{5}\}$

# Multiplicity

The **multiplicity** of a zero is the degree of its corresponding linear factor.

**Directions:** Find the zeros of the functions below and state their multiplicities.

**7.** $f(x) = x^3 + 10x^2 + 25x$

$f(x) = x(x^2 + 10x + 25)$

$f(x) = x(x+5)(x+5)$

$f(x) = x(x+5)^2$

$x = -5$; mult. of 2, $x = 0$; mult. of 1

**8.** $f(x) = -x^4 + 6x^3 - 9x^2$

$f(x) = -x^2(x^2 - 6x + 9)$

$f(x) = -x^2(x-3)^2$

$x = 0$; mult. of 2

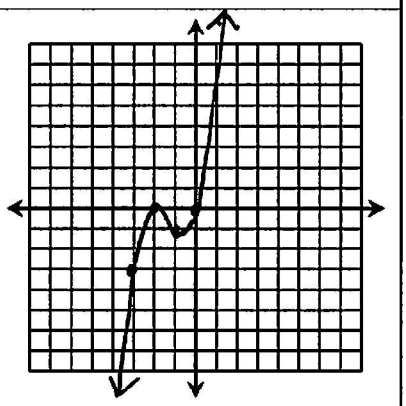
$x = 3$; mult. of 2

# Multiplicity & Graphing

Write $f(x) = x^3 + 4x^2 + 4x$ in factored form and graph the function. Then look at how the multiplicities affect the graph.

**Factored Form**

$f(x) = x(x+2)(x+2)$



- When the **multiplicity** is EVEN, the graph is tangent to the $x$-axis at that zero.
- When the **multiplicity** is ODD, the graph intersects the $x$-axis at that zero.

# More Examples

Identify the zeros, their multiplicities, and describe the effect on the graph.

**9.** $f(x) = (x-1)^2(2x-3)^2(x+4)$

| Zero | Multiplicity | Effect |
|---|---|---|
| $-4$ | 1 | intersects |
| 1 | 2 | tangent |
| $\frac{3}{2}$ | 2 | tangent |

**10.** $f(x) = x^5 + 8x^4 + 16x^3$

$f(x) = x^3(x^2 + 8x + 16)$

$f(x) = x^3(x+4)^2$

| Zero | Multiplicity | Effect |
|---|---|---|
| $-4$ | 2 | tangent |
| 0 | 3 | intersects |
| | | |

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**Date:** _____ **Per:** _____

**Unit 3:** Power, Polynomials, and Rational Functions

**Homework 4:** Zeros and Multiplicity

**** This is a 2-page document! ****

**Directions:** Find the zeros of the functions below by factoring.

**1.** $f(x)=x^3-5x^2+6x$

$f(x)=x(x^2-5x+6)$

$f(x)=x(x-2)(x-3)$

$x=\{0,2,3\}$

**2.** $f(x)=2x^3+22x^2+56x$

$f(x)=2x(x^2+11x+28)$

$f(x)=2x(x+4)(x+7)$

$x=\{-7,-4,0\}$

**3.** $f(x)=x^4-17x^2+16$

$f(x)=(x^2-16)(x^2-1)$

$f(x)=(x+4)(x-4)(x+1)(x-1)$

$x=\{\pm1,\pm4\}$

**4.** $f(x)=2x^3+5x^2-7x$

$f(x)=x(2x^2+5x-7)$

$f(x)=x(2x+7)(x-1)$

$x=\{-\frac{7}{2},0,1\}$

**5.** $f(x)=6x^3+7x^2-5x$

$f(x)=x(6x^2+7x-5)$

$f(x)=x(3x+5)(2x-1)$

$x=\{-\frac{5}{3},0,\frac{1}{2}\}$

**6.** $f(x)=-3x^3-13x^2-4x$

$f(x)=-x(3x^2+13x+4)$

$f(x)=-x(3x+1)(x+4)$

$x=\{-4,-\frac{1}{3},0\}$

**7.** $f(x)=4x^4-5x^2+1$

$f(x)=(4x^2-1)(x^2-1)$

$f(x)=(2x+1)(2x-1)(x+1)(x-1)$

$x=\{\pm\frac{1}{2},\pm1\}$

**8.** $f(x)=4x^4+80x^3+204x^2$

$f(x)=4x^2(x^2+20x+51)$

$f(x)=4x^2(x+17)(x+3)$

$x=\{-17,-3,0\}$

**9.** $f(x)=4x^3+16x^2-25x-100$

$f(x)=4x^2(x+4)-25(x+4)$

$f(x)=(4x^2-25)(x+4)$

$f(x)=(2x+5)(2x-5)(x+4)$

$x=\{-4,\pm\frac{5}{2}\}$

**10.** $f(x)=3x^3-x^2-12x+4$

$f(x)=x^2(3x-1)-4(3x-1)$

$f(x)=(x^2-4)(3x-1)$

$f(x)=(x+2)(x-2)(3x-1)$

$x=\{\frac{1}{3},\pm2\}$

**Directions:** Find the zeros of the functions below and state their multiplicities

**11.** $f(x) = x^3 + 14x^2 + 49x$

$f(x) = x(x^2+14x+49)$

$f(x) = x(x+7)^2$

$x=-7$; mult. of 2
$x=0$; mult. of 1

**12.** $f(x) = x^4 + 18x^3 + 81x^2$

$f(x) = x^2(x^2+18x+81)$

$f(x) = x^2(x+9)^2$

$x=-9$; mult. of 2
$x=0$; mult. of 2

**13.** $f(x) = 4x^3 - 4x^2 + x$

$f(x) = x(4x^2-4x+1)$

$f(x) = x(2x-1)^2$

$x=0$; mult. of 1
$x=\frac{1}{2}$; mult. of 2

**14.** $f(x) = x^3 + 5x^2 - x - 5$

$f(x) = x^2(x+5) - 1(x+5)$

$f(x) = (x^2-1)(x+5)$

$f(x) = (x+1)(x-1)(x+5)$

$x=-5$; mult. of 1
$x=-1$; mult. of 1 $x=1$; mult. of 1

**15.** $f(x) = 36x^4 - 12x^3 + x^2$

$f(x) = x^2(36x^2-12x+1)$

$f(x) = x^2(6x-1)^2$

$x=0$; mult. of 2
$x=\frac{1}{6}$; mult. of 2

**16.** $f(x) = x^5 + 16x^4 + 64x^3$

$f(x) = x^3(x^2+16x+64)$

$f(x) = x^3(x+8)^2$

$x=-8$; Mult. of 2
$x=0$; Mult. of 3

**Directions:** Identify the zeros, their multiplicities, and describe the effect on the graph.

**17.** $f(x) = (x-2)^2(3x+7)^3(4x-3)^2$

| Zero | Multiplicity | Effect |
|---|---|---|
| $-\frac{7}{3}$ | 3 | intersects |
| $\frac{3}{4}$ | 2 | tangent |
| 2 | 2 | tangent |

**18.** $f(x) = x^2(x-9)(2x+5)^2$

| Zero | Multiplicity | Effect |
|---|---|---|
| $-\frac{5}{2}$ | 2 | tangent |
| 0 | 2 | tangent |
| 9 | 1 | intersects |

**19.** $f(x) = x^5 + 6x^4 + 9x^3$

$f(x) = x^3(x^2+6x+9)$

$f(x) = x^3(x+3)^2$

| Zero | Multiplicity | Effect |
|---|---|---|
| -3 | 2 | tangent |
| 0 | 3 | intersects |
| | | |

**20.** $f(x) = 8x^3 + 24x^2 + 18x$

$f(x) = 2x(4x^2+12x+9)$

$f(x) = 2x(2x+3)^2$

| Zero | Multiplicity | Effect |
|---|---|---|
| $-\frac{3}{2}$ | 2 | tangent |
| 0 | 1 | intersects |
| | | |